

13.a. Perform ADF test by describing the data as an AR(2) model with OLS estimation. 16 4 3 4

Year	1	2	3	4	5	6	7	8	9	10
Sales (units)	20	21	23	22	25	24	24	27	26	28

(OR)

b. Compute the correlation, ACF, ACVF, PACF values for the following data and draw the plotting. 16 4 3 4

Year	1	2	3	4	5	6	7	8	9	10
Sales (units)	5.2	4.9	5.5	4.9	5.2	5.7	5.4	5.8	5.9	6

14.a.i. Discuss the causality invertibility of the model. 8 3 4 2
 $x_t = 0.44x_{t-1} + 0.45x_{t-2} + \varepsilon_t + \varepsilon_{t-1} + 0.2\varepsilon_{t-2}$

ii. Describe briefly about SARIMA model. 8 3 4 2

(OR)

b. Describe ARIMA model with examples. 16 3 5 2

15.a. Write detailed note on VAR model. 16 3 5 2

(OR)

b. Write detailed description on VARMA model. 16 3 5 2

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